

Empirical Research in Finance

5. Time series and panel data

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Outline

Time-series models

Panel data

- Pooled OLS

- Fixed effects

- Two-way FE

- Standard errors

- A finance example

Random vs. Fixed Effects

Wrap-up

Time series — the basics

- ▶ A time series tracks one entity repeatedly over time: stock prices, GDP, the VIX, oil prices, Apple's quarterly earnings.
- ▶ The data are ordered — Q2 follows Q1 — and the past affects the future, not the reverse.
- ▶ Today we only need two ideas from time series:
 1. Calendar dummies (quarter, year) are the building block of *time fixed effects* in panel data.
 2. Many series *trend* over time, and ignoring the trend produces spurious correlations.

Simple time-series models

► **Static model:** $y_t = \beta_0 + \beta_1 x_{1t} + \cdots + u_t$

► **Finite distributed lag model:**

$$y_t = \alpha + \delta_0 x_t + \delta_1 x_{t-1} + \delta_2 x_{t-2} + \cdots + u_t$$

(a change in x may take time to affect y)

► **Model with a trend:** $y_t = \alpha + \beta x_t + \gamma t + u_t$

► **Model with seasonality:**

$$y_t = \alpha + \beta x_t + \delta_2 Q_2 + \delta_3 Q_3 + \delta_4 Q_4 + u_t$$

Note: the quarterly dummies Q_2, Q_3, Q_4 are exactly the same idea as the *time fixed effects* we will add to panel regressions later — one dummy per period, with one period dropped as the baseline.

The trend trap — HSEINV (Wooldridge 10.7)

Annual U.S. data:

- ▶ `linvpc`: log of per-capita housing investment (= construction of new houses)
- ▶ `lprice`: log of housing price index

Static model:

$$\log y_t = \beta_0 + \beta_1 \log x_t + u_t.$$

If both housing investment and prices have trended up over 40 years, would you trust $\hat{\beta}_1$ from this regression?

HSEINV — regression #1

```
> require(wooldridge); attach(hseinv)
> m <- lm(linvpc ~ lprice); summary(m)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-0.55023	0.04303	-12.788	1.03e-15	***
lprice	1.24094	0.38242	3.245	0.00238	**

Residual standard error: 0.1554 on 40 d.f.
Multiple R-squared: 0.2084

$\hat{\beta}_1 = +1.24$. “A 1% rise in house prices raises construction by 1.24%.” Plausible? Or are both variables just drifting up together?

HSEINV — regression #2 (add a time trend)

	Estimate	Std. Error	<i>p</i> -value
lprice	− 0.381	0.679	0.578
t	+0.0098	0.0035	0.008

Once we control for the trend t :

- ▶ the price coefficient *collapses to zero* (and changes sign),
- ▶ the trend t takes over: $\approx 1\%$ growth in investment per year.

The original “effect” was the trend in disguise.

Why the coefficient flipped

- ▶ Both `lnvpc` and `lprice` drift upward over the sample.
- ▶ Without t , OLS attributes the shared upward drift to `lprice`.
- ▶ With t in the regression, the drift is absorbed by t , and what remains is the *deviation* of each variable from its trend.
- ▶ **Lesson:** trending variables can produce *spurious* statistical significance. Always ask whether the variation in your right-hand side is between-trend or around-the-trend.

Panel Data - An example: fertilizer on plots

Effect of fertilizer on crop yield. We send an assistant to put random amounts of fertilizer on n different plots over T years.

plot	year	yield (kg)	fertilizer (kg)
1	1	68.81262	14.20747
1	2	67.82115	11.11035
1	3	78.37679	25.28446
1	4	78.12128	15.75408
2	1	63.78051	32.00421
2	2	64.87444	31.98836
2	3	60.46330	20.08297
2	4	65.30558	23.98785
3	1	59.61762	48.17068
3	2	57.02471	43.02786
3	3	56.48006	42.99498
3	4	62.16982	42.44634
...			

Panel structure: $plot(i) \times year(t)$.

Starting point: pooled OLS

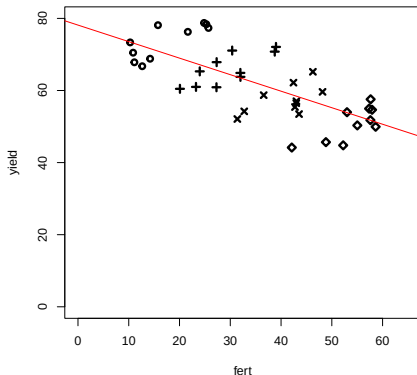
$$\text{yield}_{it} = \alpha + \beta \text{fert}_{it} + u_{it}.$$

```
> yielddat <- read.csv("additional/05-yields.csv")  
> m <- lm(yield ~ fert, data = yielddat)
```

	Estimate	Std. Error	p-value
(Intercept)	78.14	2.88	< 0.001
fert	− 0.46	0.07	< 0.001

Fertilizer *reduces* yield? Look at the picture before believing the number.

Pooled OLS, graphically



The negative slope is *between* plots. Within each plot, more fertilizer goes with more yield.

The story behind the sign

- ▶ The assistant put more fertilizer on the worst plots.
- ▶ Plot quality α_j is unobserved but constant over time.
- ▶ Decompose the error: $u_{it} = \alpha_j + v_{it}$.

Add a dummy for each plot:

$$\text{yield}_{it} = \underbrace{\alpha_j}_{\text{plot fixed effect}} + \beta \text{fert}_{it} + v_{it}.$$

Fixed effects absorb *unobserved time-invariant heterogeneity*.

```
> m <- lm(yield ~ fert + factor(plot) - 1, data = yielddat)
```

Code dictionary: `factor(plot)` = one dummy per plot; `- 1` drops the intercept so each plot dummy gets its own coefficient.

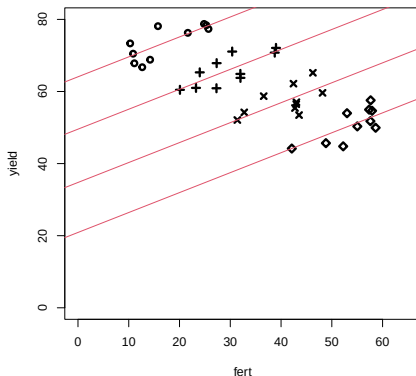
One-way FE result

	Estimate	Std. Error	<i>p</i> -value
fert	+0.55	0.09	< 0.001
factor(plot)1	64.08	1.81	< 0.001
...	...	...	...

- One row per plot is estimated (4 plot dummies); only one shown.

The coefficient on fertilizer flipped: $-0.46 \rightarrow +0.55$.

One-way FE, graphically



Each plot now has its own intercept; β is identified from *within-plot* variation.

Add year FE: weather changes year to year

$$\text{yield}_{it} = \underbrace{\alpha_i}_{\text{plot FE}} + \underbrace{\eta_t}_{\text{year FE}} + \beta \text{fert}_{it} + \varepsilon_{it}.$$

- ▶ Plot FE absorbs anything fixed about the plot.
- ▶ Year FE absorbs anything common to all plots in that year (weather, prices).
- ▶ Together = “two-way fixed effects”.

```
> m <- lm(yield ~ fert + factor(plot) + factor(year) - 1,  
+         data = yielddat)
```

Two-way FE result

	Estimate	Std. Error	<i>p</i> -value
fert	+ 0.56	0.03	< 0.001
factor(plot)1	61.51	0.90	< 0.001
factor(year)4	6.51	0.80	< 0.001
...	...	...	...

- ▶ Plot *and* year dummies are now estimated (4 plots + 9 years); only one of each shown.
- ▶ Often, we don't read the dummy coefficients: their job is to soak up unwanted variation, not to be interpreted.

Same point estimate as one-way FE, but a tighter SE — adding year FE removed year-to-year noise from β 's denominator.

Estimating without printing every dummy: plm

```
> require(plm)
> pm <- plm(yield ~ fert,
+           data = yieldddat,
+           index = c("plot","year"),
+           model = "within")
> summary(pm)
```

	Estimate	Std. Error	<i>p</i> -value
fert	+0.56	0.088	< 0.001

- Add factor(year) to get two-way FE.

Standard errors: why they are wrong — and what to do

Default OLS SEs assume errors are (i) identically distributed and (ii) independent across observations. Both fail in panels :

- ▶ **Heteroskedasticity** (variance differs across observations) $\Rightarrow$ *robust* SEs.
- ▶ **Within-cluster correlation** (errors correlated within firm across years; or across firms in the same year) $\Rightarrow$ *clustered* SEs.

Three common cluster choices:

- ▶ **By individual** (firm/bank/borrower) — within-firm correlation.
- ▶ **By time** (year/quarter) — within-year correlation across firms.
- ▶ **Two-way** — both at once.

Pick the cluster level that matches the dependence in your data, not the one that gives the prettiest t -stat.

Standard errors: practical rules

Robust (heteroskedasticity only)

- ▶ Does *not* fix within-group correlation.

Clustered (heteroskedasticity + within-group correlation)

- ▶ Preferred whenever errors might be correlated within firm or within year.
- ▶ SEs can go *up or down* after clustering — direction is empirical, not theoretical.
- ▶ Need $\gtrsim 30$ clusters.

One-line implementation (cluster by firm):

R

```
coeftest(pm, vcov = vcovHC(pm, cluster = "group"))
```

* Stata

```
xtreg y x, fe vce(cluster firm)
```

Python

```
PanelOLS(...).fit(cov_type="clustered", cluster_entity=True)
```

Full reference in the appendix.

What does a finance panel look like?

Compustat-style extract (toy, 4 firms $\times$ 3 years):

firm	year	leverage	ROA	log(assets)
AAPL	2018	0.31	0.16	12.6
AAPL	2019	0.33	0.16	12.7
AAPL	2020	0.36	0.18	12.8
MSFT	2018	0.20	0.15	12.5
MSFT	2019	0.22	0.16	12.5
MSFT	2020	0.21	0.18	12.6
GE	2018	0.55	0.04	13.0
GE	2019	0.58	0.05	13.0
GE	2020	0.61	0.03	12.9
F	2018	0.62	0.03	12.4
...				

Two indices: *firm* (i) and *year* (t). Every observation is a firm-year.

Worked example — the question

Do more profitable firms borrow more, or less?

Two competing finance theories give opposite predictions:

- ▶ **Pecking order** (Myers, 1984): firms prefer internal funds first, then debt, then equity.
 - ▶ More profitable $\Rightarrow$ more internal cash $\Rightarrow$ *less* debt.
- ▶ **Trade-off theory**: firms balance tax shields against bankruptcy costs.
 - ▶ More profitable $\Rightarrow$ safer $\Rightarrow$ more debt capacity $\Rightarrow$ *more* debt.

Which sign do you expect the regression coefficient on profitability to have?

Step 1 — pooled OLS

Use a Compustat panel: firm-year, $\sim 2,000$ firms over 20 years.

$$\text{lev}_{it} = \alpha + \beta \text{ROA}_{it} + \varepsilon_{it}.$$

```
> m <- lm(lev ~ ROA, data = compustat)
```

	Estimate	Cluster SE	t-stat
ROA	-0.18	(0.04)	[-4.5]

Reading: “On average, more profitable firms have lower leverage.”
Looks like pecking order. *But* we are comparing Apple to General Motors to a tiny biotech — they differ in many unobserved ways (managerial talent, corporate culture, quality of bank relationships).

Step 2 — absorb firm differences (firm FE)

Re-ask the question *within* each firm:

$$\text{lev}_{it} = \alpha_i + \beta \text{ROA}_{it} + \varepsilon_{it}.$$

Firm FE absorbs anything constant about the firm — both observable (industry, HQ country) and unobservable (managerial talent, corporate culture, bank relationships).

Now β answers: *when a given firm becomes more profitable, does its own leverage change?*

```
> pm <- plm(lev ~ ROA, data = compustat,  
+          index = c("gvkey","fyear"), model = "within")
```

	Estimate	Cluster SE	t-stat
ROA	-0.42	(0.06)	[-7.0]

The effect more than doubles — within firms, profitability and debt move sharply in opposite directions.

Step 3 — absorb the macro environment (year FE)

Add a year fixed effect to net out shocks common to all firms in a given year:

$$\text{lev}_{it} = \alpha_i + \eta_t + \beta \text{ROA}_{it} + \varepsilon_{it}.$$

Year FE absorbs: interest rates, recessions, tax reforms, financial crises, regulation.

```
> pm <- plm(lev ~ ROA, data = compustat,  
+           index = c("gvkey","fyear"),  
+           model  = "within", effect = "twoways")  
> coeftest(pm, vcov = vcovHC(pm, cluster = "group"))
```

	Estimate	Cluster SE	t-stat
ROA	-0.46	(0.06)	[-7.7]

The result barely moves — it is robust to controlling for the macro environment.

Step 4 — the three columns side by side

	Pooled OLS	Firm FE	Firm + Year FE
ROA (coefficient)	-0.18	-0.42	-0.46
t-statistic	[-4.5]	[-7.0]	[-7.7]
Firm FE	no	yes	yes
Year FE	no	no	yes
Cluster	firm	firm	firm

- ▶ Sign is the same in all three columns.
- ▶ Pooled OLS *understates* the within-firm effect by more than half.
- ▶ The pooled estimate was diluted by cross-firm noise; firm FE strips it out.

What to take away

- ▶ **Always run all three columns.** The *gap* between pooled OLS and FE is informative — it tells you how much of the raw correlation comes from cross-firm differences vs. within-firm dynamics.
- ▶ **Use the most demanding specification as baseline** (here: firm + year FE), and use the simpler ones to show readers what changed.
- ▶ **Cluster at the level of the FE that absorbs the most variation** (here: firm).
- ▶ **Connect the result back to theory.** Here, the data is consistent with pecking order at the within-firm level; trade-off theory predictions would need extra structure (e.g. deviations from a target leverage).

Random Effects vs. Fixed Effects

- ▶ **FE assumption:** α_i may be correlated with the regressors.
 - ▶ *Fertilizer:* more fertilizer goes to worse plots $\Rightarrow \text{Cov}(\alpha_i, \text{fert}_{it}) \neq 0$.
 - ▶ *Finance:* a firm's unobserved quality α_i is correlated with its profitability $\Rightarrow \text{Cov}(\alpha_i, \text{ROA}_{it}) \neq 0$.
- ▶ **RE assumption:** α_i is uncorrelated with the regressors.
 - ▶ *Fertilizer:* fertilizer is randomly assigned $\Rightarrow \text{Cov}(\alpha_i, \text{fert}_{it}) = 0$.
 - ▶ *Finance:* requires (e.g.) a randomised regulatory experiment, where the variable of interest is independent of unobserved firm quality.
Rare.
- ▶ Random assignment is the exception in finance. Use FE unless you have a strong design argument.

$\Rightarrow$ In finance applications, FE is the default; RE shows up mostly in classical labour-econ panels.

Checkpoint – try these yourself

Example 3: Executive compensation and firm size

A pooled-OLS regression of executive pay on firm size gives $\hat{\beta}_{\text{pooled}} = +0.40$. Once firm FE is added, $\hat{\beta}_{\text{firm FE}} = +0.10$.

1. What unobserved firm characteristic was inflating the pooled-OLS estimate?
2. What is $+0.10$ telling us that $+0.40$ was not?

Example 4: Headquartered in California

You want to estimate the effect of being headquartered in California on a firm's R&D spending, using a panel of US firms over 10 years.

1. Why is *firm FE* not useful here?
2. What would you use instead?

Reading

- ▶ Wooldridge, ch. 10 — time series, trends, seasonality
- ▶ Wooldridge, ch. 13–14 — panel data and fixed effects

Appendix A — Fixed-effects regression

Notation in the code below: `firm` is the cross-sectional unit (replace with whatever your entity column is called — `bank`, `fund`, etc.), `year` is the time column. In the lecture math notation these were i and t .

	Textbook	Modern
R	<code>plm(y ~ x, data, index=c("firm","year"), model="within")</code>	<code>feols(y ~ x firm, data)</code>
Stata	<code>xtset firm year; xtreg y x, fe</code>	<code>reghdfe y x, absorb(firm)</code>
Python	<code>PanelOLS.from_formula("y ~ x + EntityEffects", df).fit()</code>	<code>feols("y ~ x firm", df)</code>

Two-way FE (add the time dimension):

- ▶ Textbook: `+ factor(year)` (R) / `i.year` (Stata) / `+ TimeEffects` (Python)
- ▶ Modern: `| firm + year (fixest)` / `absorb(firm year)` (`reghdfe`) / `| firm + year` (`pyfixest`)

Appendix B — Clustering standard errors

Notation in the code: `firm` is the cross-sectional column, `year` is the time column (rename to your own). Cluster-by-firm shown below.

	Textbook	Modern
R	<code>coeftest(pm, vcov = vcovHC(pm, cluster = "group"))</code>	<code>feols(y ~ x firm, cluster = ~firm, data)</code>
Stata	<code>xtreg y x, fe vce(cluster firm)</code>	<code>reghdfe y x, absorb(firm) vce(cluster firm)</code>
Python	<code>PanelOLS(...).fit(cov_type="clustered", cluster_entity=True)</code>	<code>feols("y ~ x firm", df, vcov={"CRV1":"firm"})</code>

Variants:

- ▶ *Cluster by year*: swap `firm` for `year` in each cell.
- ▶ *Two-way cluster*: `cluster = ~firm + ~year (fixest); vce(cluster firm#year) (reghdfe)`.